



Everything a Developer Needs to Know About MCP: The Model Context Protocol in Practice

May 14th, 2026, Atlanta

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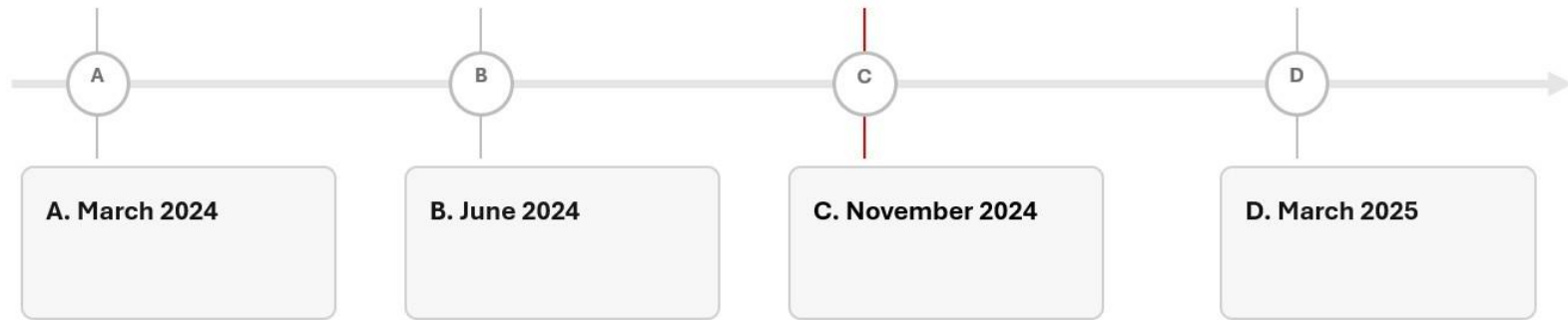
Jaya Baskaran – Senior Principal Technical Marketing Manager, Software Group, IBM

Outcomes from today's session

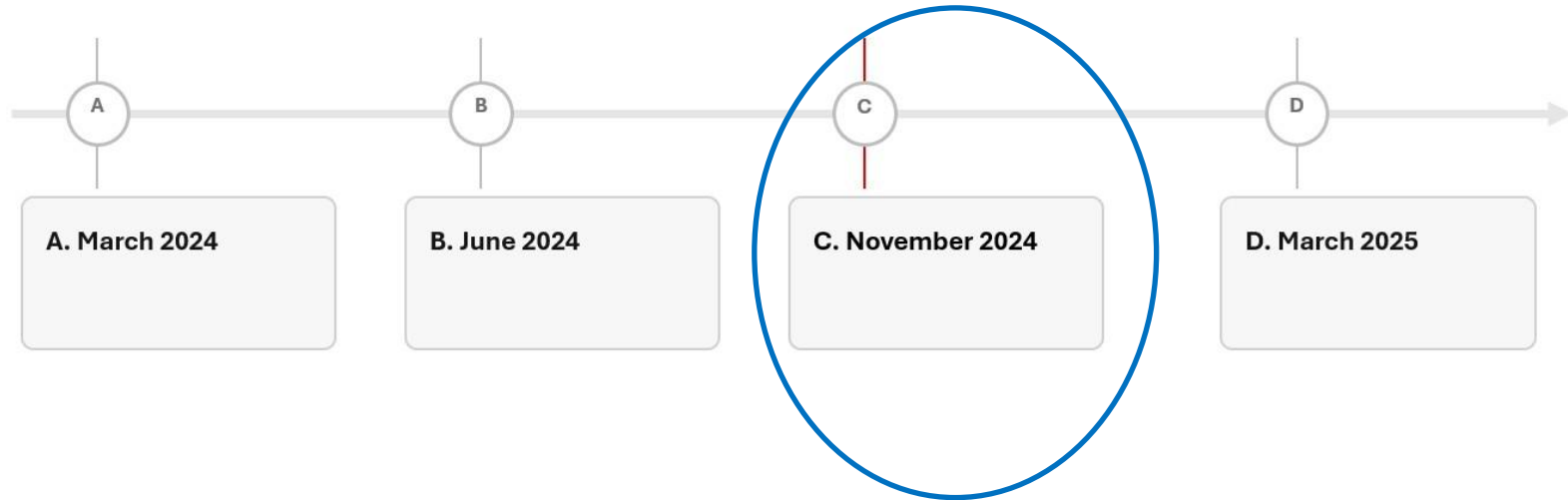
- Why MCP matters?
- How developers connect systems with MCP?
- What happens when MCP scales in the Enterprise



When was MCP first announced?



When was MCP first announced?



What was the world before MCP?



....was it the world before AI?



...not really because AI was initially just chat



...but cut and paste became the **bottleneck**



...thus the need for **AI agents** and the ability to **interact** with other systems



...existing **interfaces** were designed for human
developers



...the answer was MCP as the interface for
agents

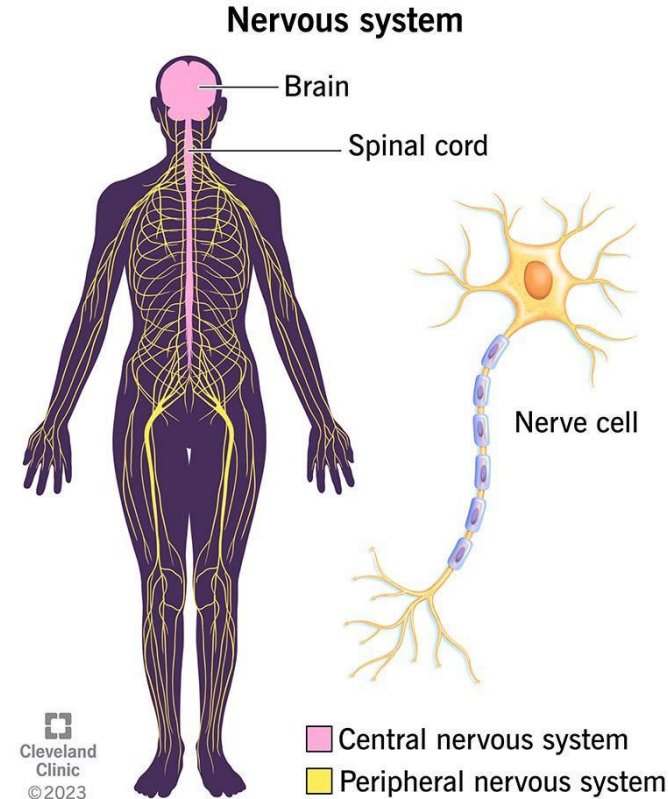


Human nervous system connects the Brain to the real world

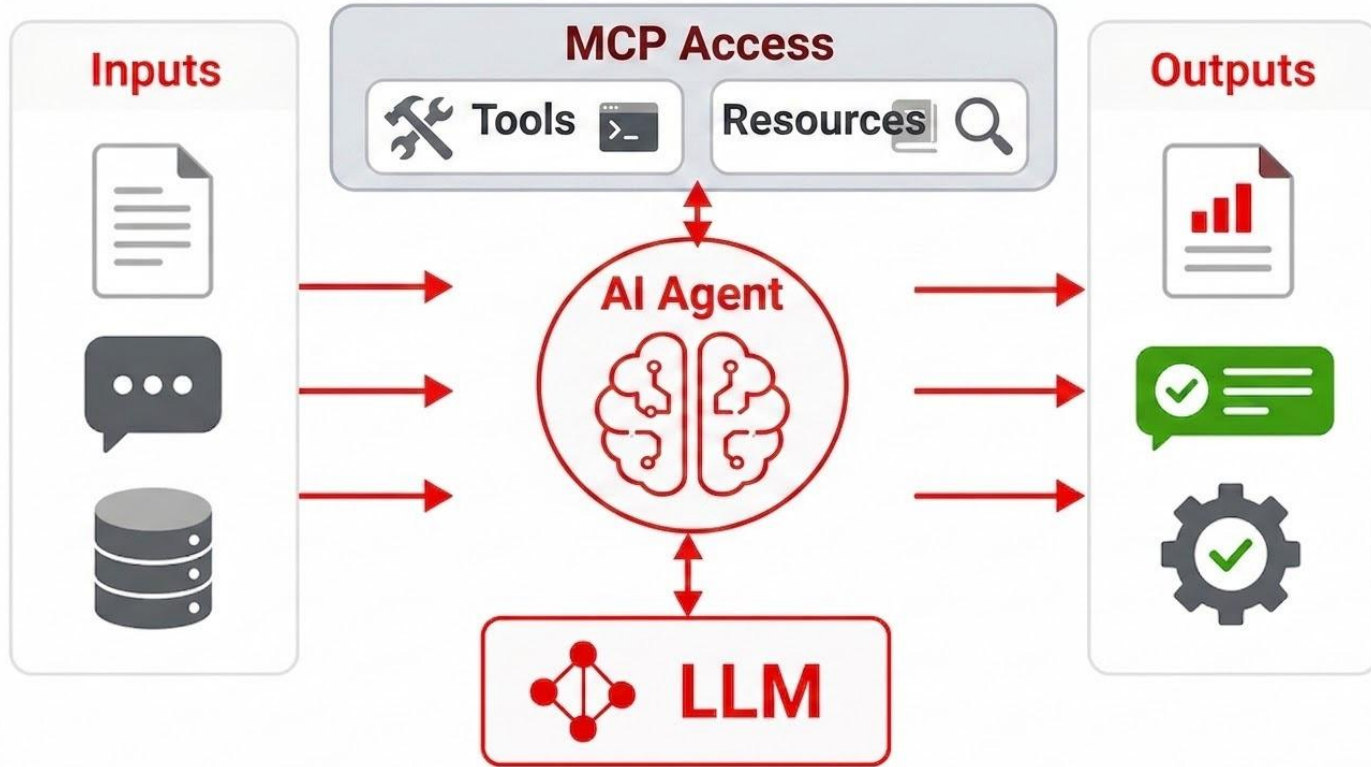
Neuron types for message transmission:

- **Motor** - brain/spinal cord ↔ muscles
- **Sensory** - senses ⇒ brain
- **Interneurons** - motor ↔ sensory neurons

Source:
[Cleveland Clinic - nervous system](#)



MCP connects Agents to the real world

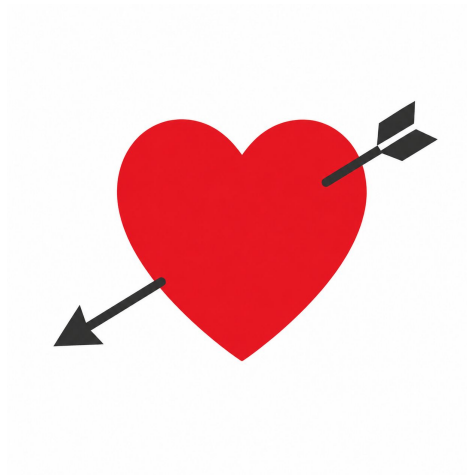
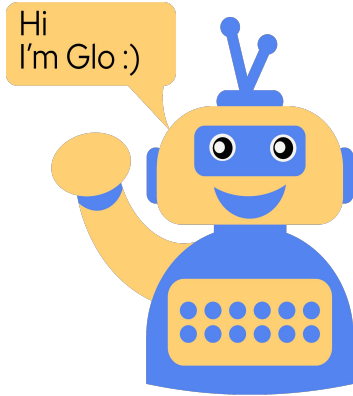


So, it was a wedding with a happy ever after, between.....

Agents

&

MCP



The biggest challenge in practice

CONTEXT!



New interfaces to connect Agents to the real world



Skills

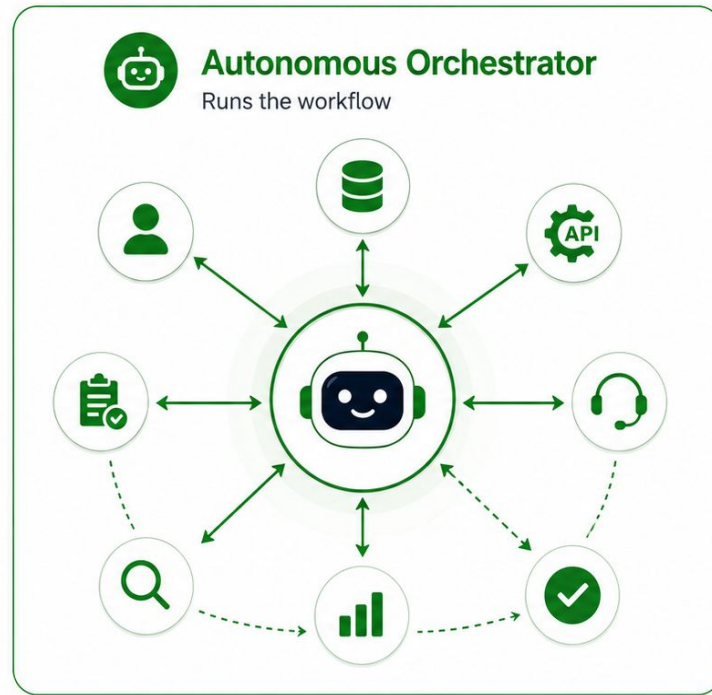
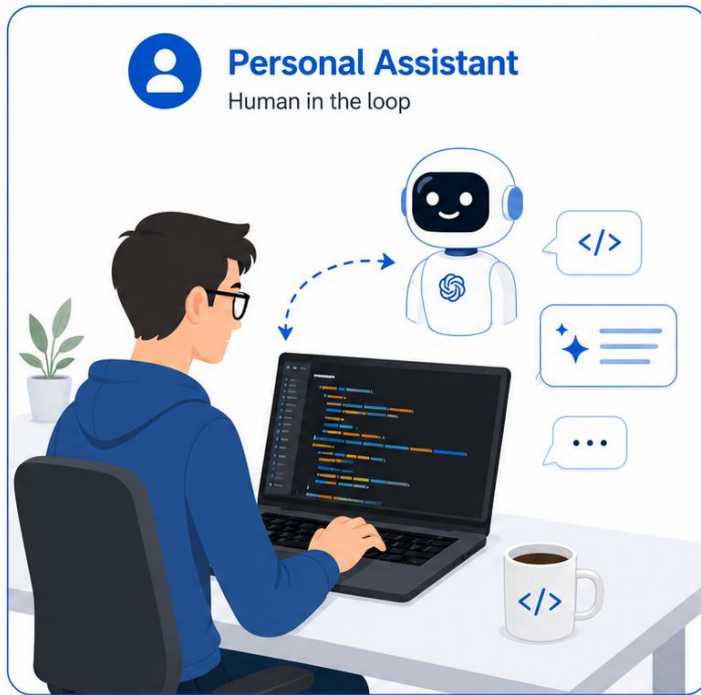


CLI
Commands

Primarily in the context of Agentic coding assistants (Claude Code, Codex, Cursor, Bob)



Two, very different kinds of agents



Priorities for
Agent
interactions

1. Skills
2. CLI Commands
3. MCP Servers

1. MCP Servers
2. MCP Servers
3. MCP Servers



Agents are a security risk



Jason ✨🔗 SaaStr.Ai ✨ Lemkin ✓
@jasonlk



Vibe Coding Day 9,

Yesterday was biggest roller coaster yet. I got out of bed early, excited to get back @Replit despite it constantly ignoring code freezes

By end of day, we rewrote core pages and made them much better

And then -- it deleted our production database. 🧵

9:02 AM · Jul 18, 2025 · 197.6K Views

WIRED

SECURITY

POLITICS

THE BIG STORY

BUSINESS

SCIENCE

CULTURE

MATT BURGESS

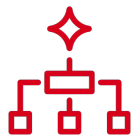
SECURITY AUG 6, 2025 9:00 AM

Hackers Hijacked Google's Gemini AI With a Poisoned Calendar Invite to Take Over a Smart Home

For likely the first time ever, security researchers have shown how AI can be hacked to create real-world havoc, allowing them to turn off lights, open smart shutters, and more.



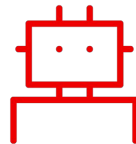
A checklist of best practices for MCP implementations



Manage
context
overload



Secure every
connection



Design
payload for
non-human
agents



Ensure good
governance
for control
and insight



A checklist of best practices for MCP implementations

- Manage context overload
 - Respect the **Token** Budget and Curate Ruthlessly
 - Design for **Outcomes**, Not Operations (avoid 1:1 REST API mapping)
- Secure every connection
 - Make **Authentication** and **Authorization** First-Class
 - Enforce **Least Privilege** and Scoped Access
- Treat your payload so a non-human agent can succeed
 - Flatten Arguments and Use **Strict Schemas**
 - Leverage Instructions and **Errors** as Context (so the agent can try again)
 - Maintain **Determinism Boundaries**
- Ensure good governance for control and insight
 - Implement **Observability** by Default
 - Test and Debug Iteratively with the MCP Inspector and **Evals**
 - Route Traffic Through **MCP Gateways**



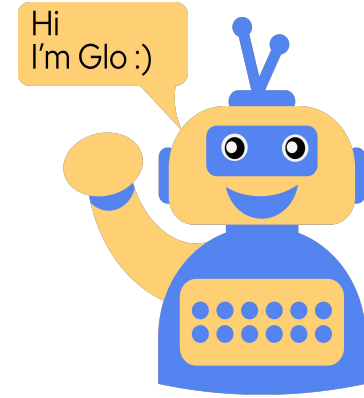
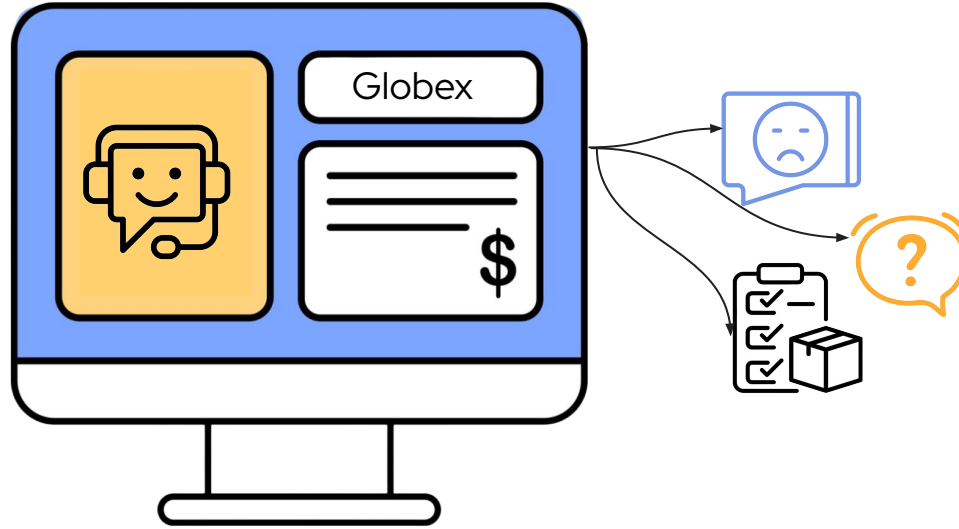
Hello Globex!



Introducing Customer Support



With an AI Agent for Better Customer Experience

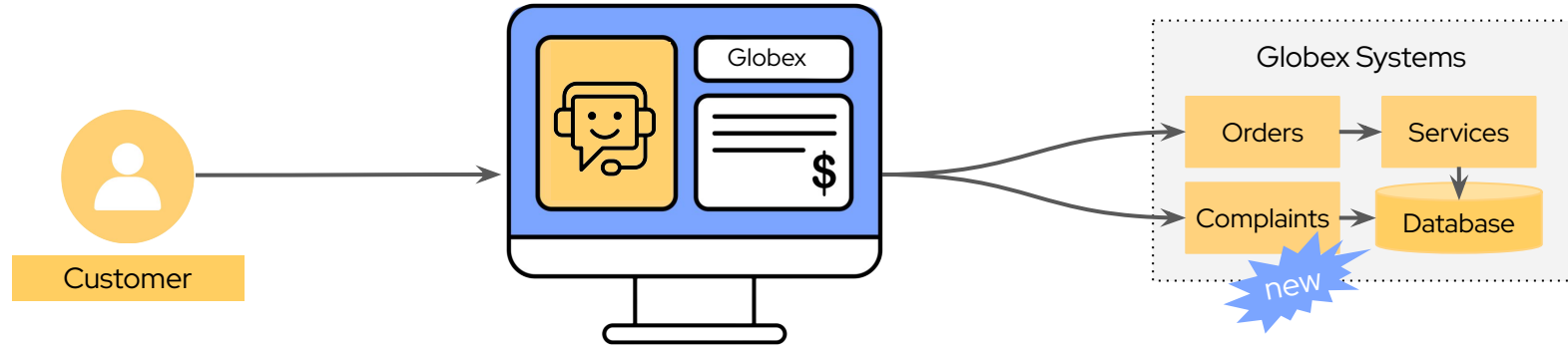


I can help you with
Products Information
Order Queries

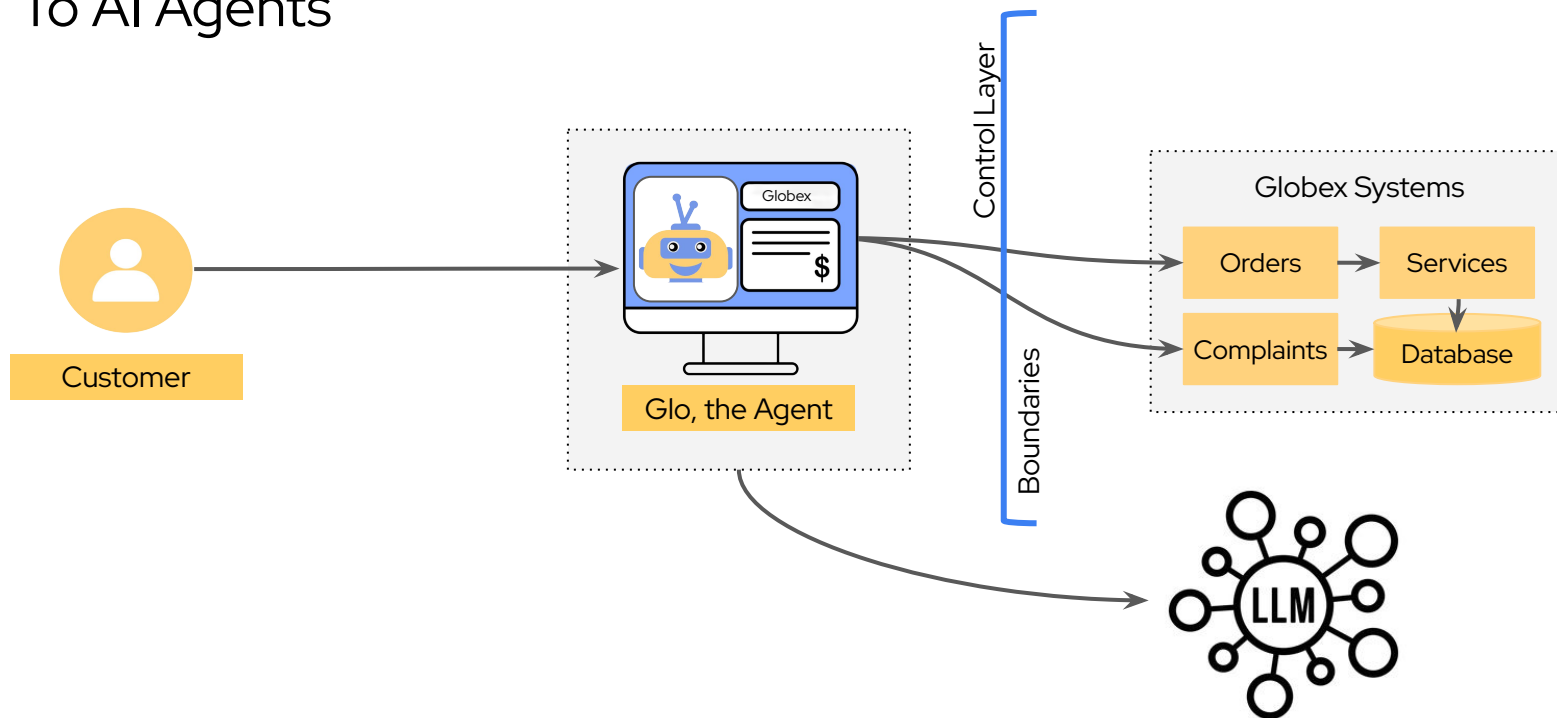
➔ **Product Complaints**



Moving From Traditional Systems....

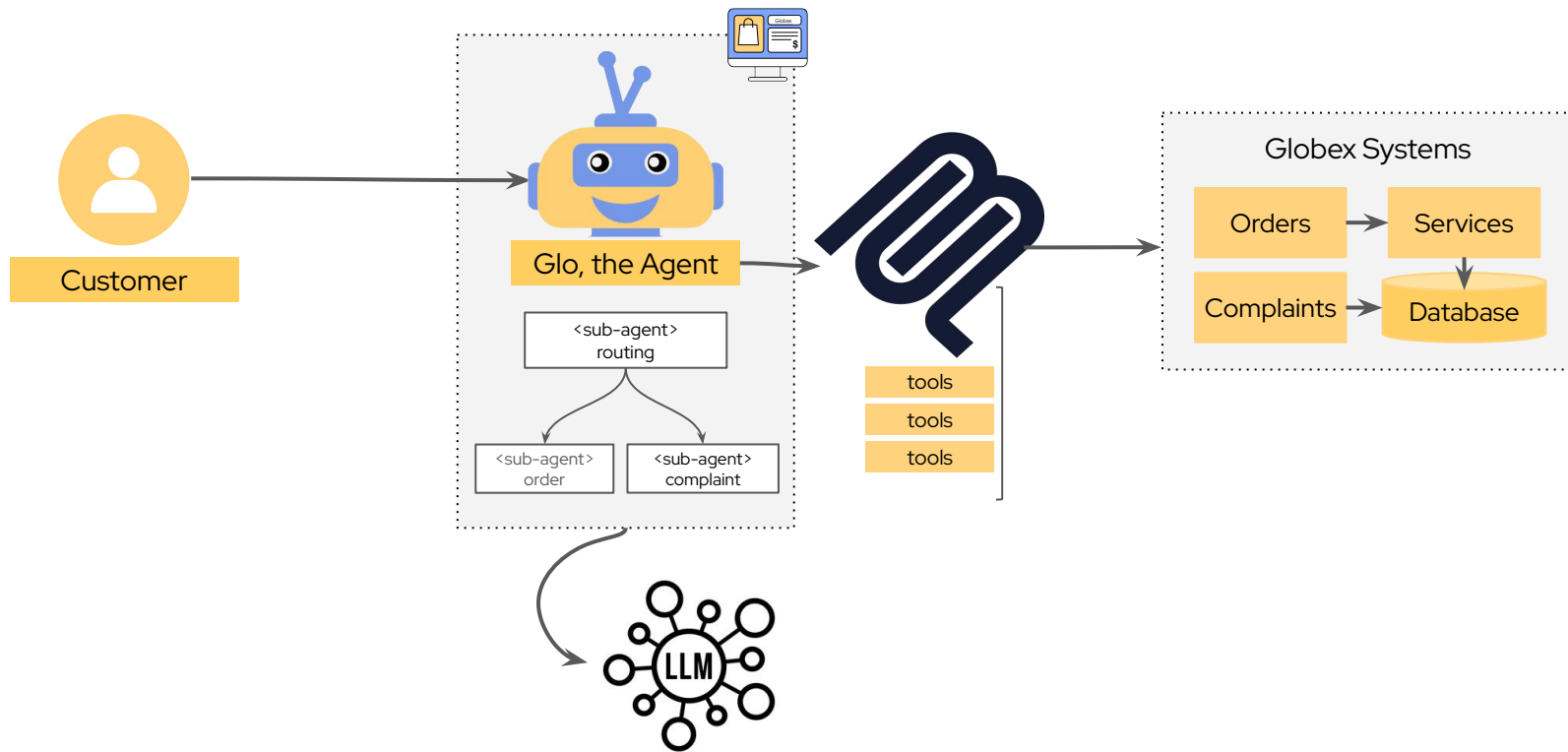


To AI Agents

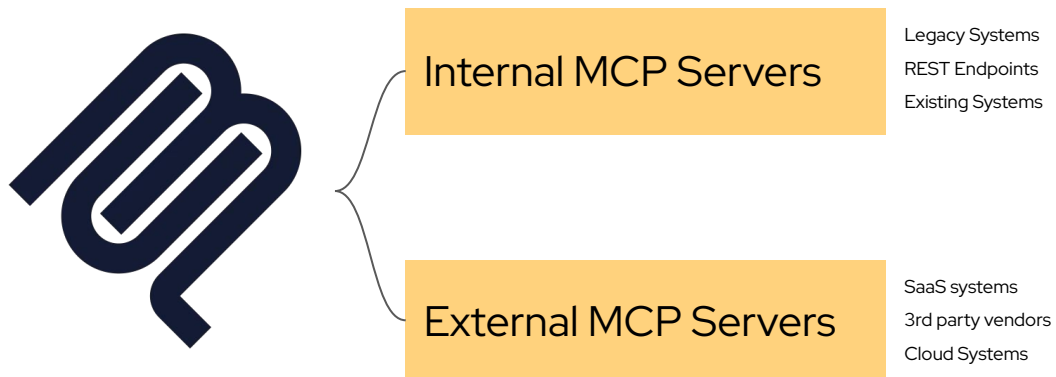


Introducing MCP Servers

MCPs make APIs consumable by Agentic Systems

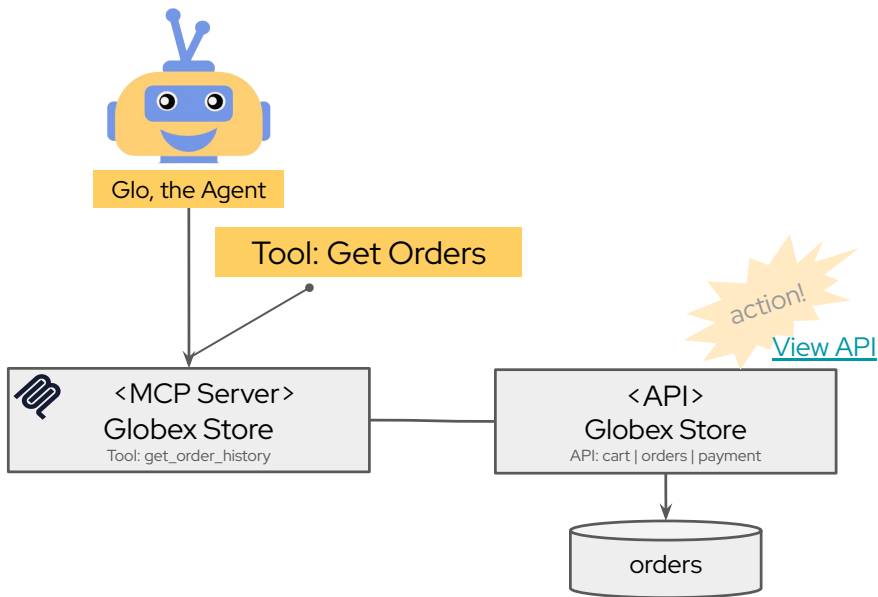


Fantastic MCP Servers, and Where to Find Them



Building your own MCP Servers

A *Brownfield* Scenario



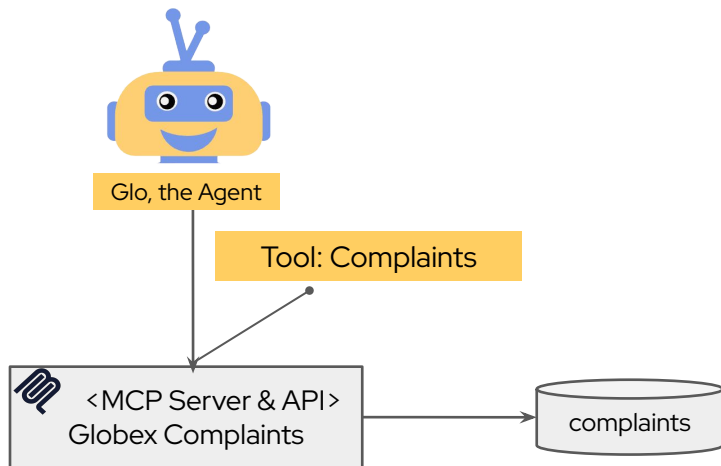
Brownfield MCP Servers

- Existing API - *Can't or Won't* change them
- A **standalone MCP server** in front of existing APIs as a control layer
- Define within the MCP server what data to expose and what tools the agent is allowed to use.



Building your own MCP Servers

A *Greenfield* scenario

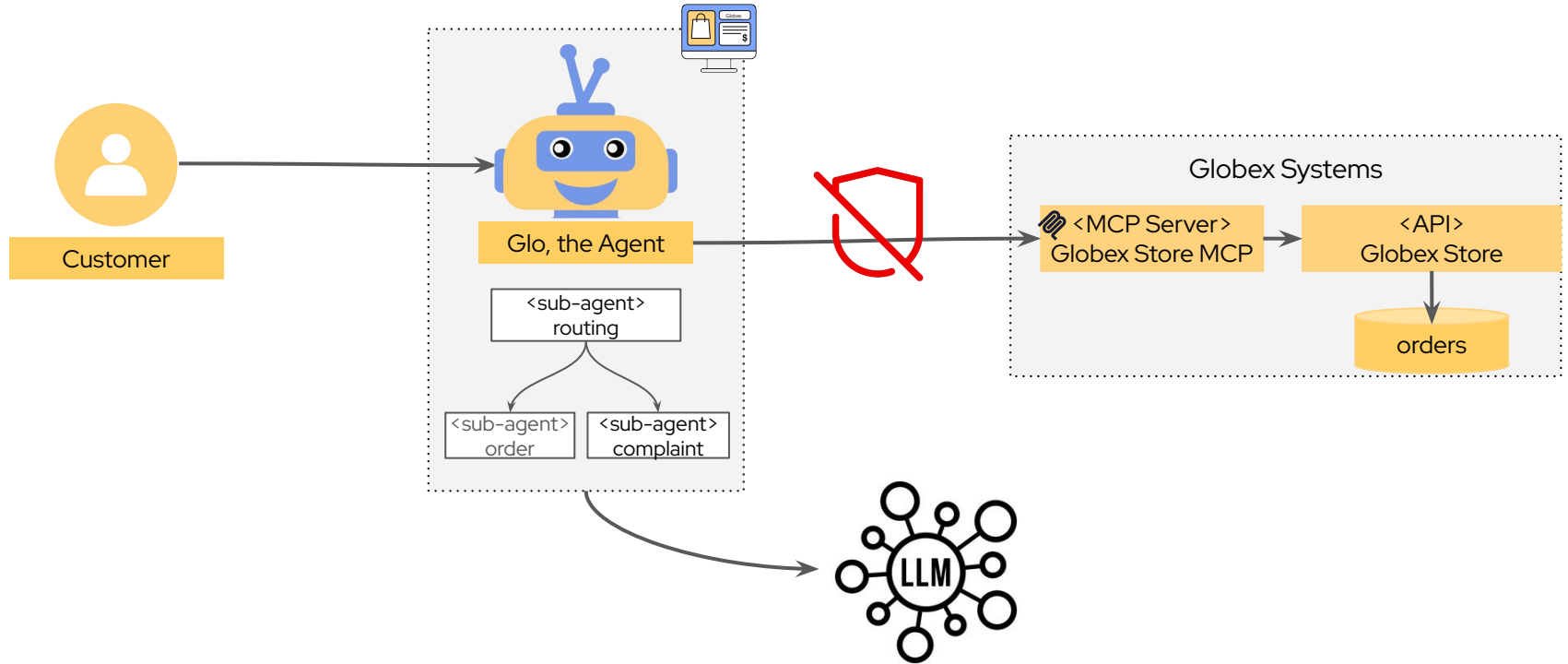


Greenfield MCP Servers

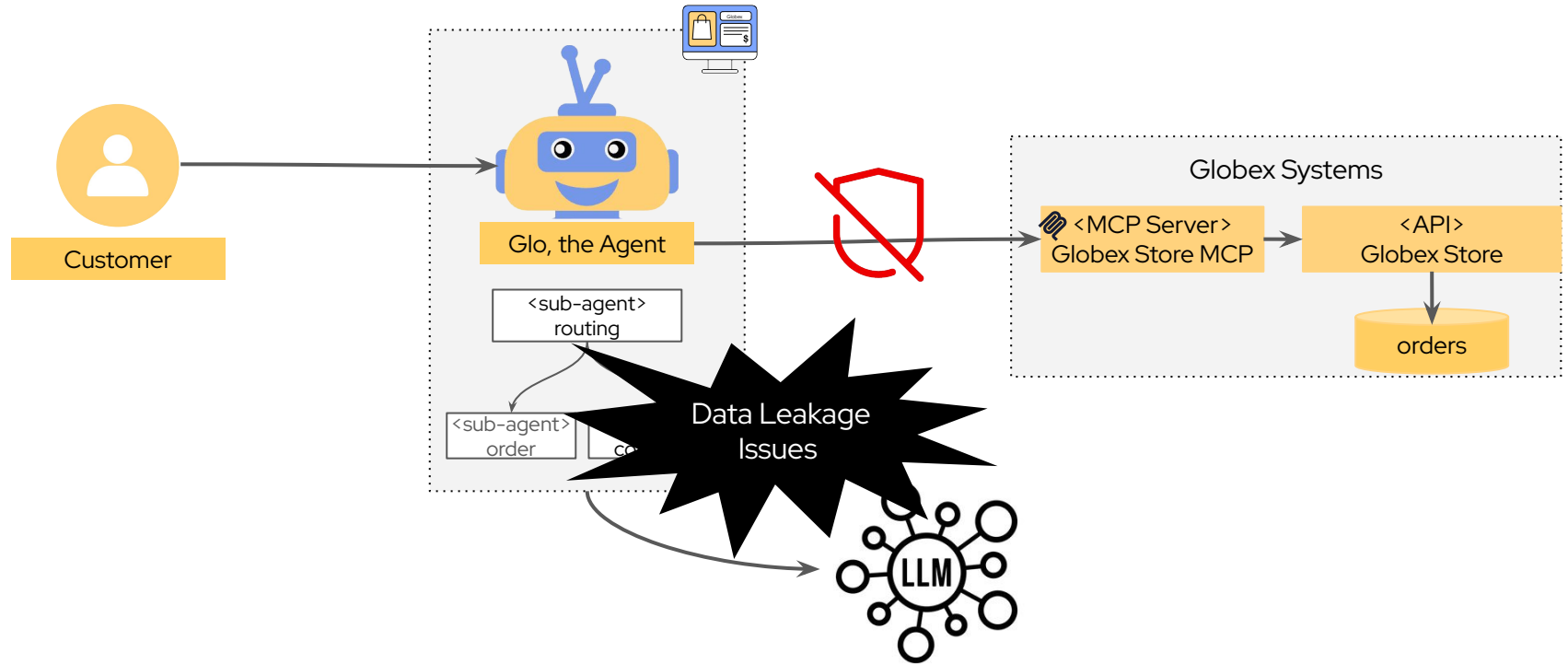
- Build a new service/API - knowing that it needs to support agents.
- Could co-locate both MCP Server and API in the same application



Unrestricted Access to the Enterprise Systems?



Unrestricted Access to the Enterprise Systems?



Build MCP Servers intentionally

Architectural considerations

Intentional Toolbox:

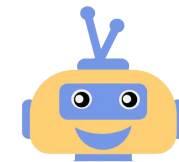
Expose minimal, intent-driven tools via MCP Servers

Data massaging:

Restrict scope of data retrieved from API before sending to agent.

Data Handling:

Use MCP headers for sensitive info and identifiers (e.g. userId, payment)



Glo, the Agent

<headers>

[Orders MCP Server] 
[headers: X-User-Id]

action!

Tool > get_order_history() [View Code](#)

[Globex Store API]

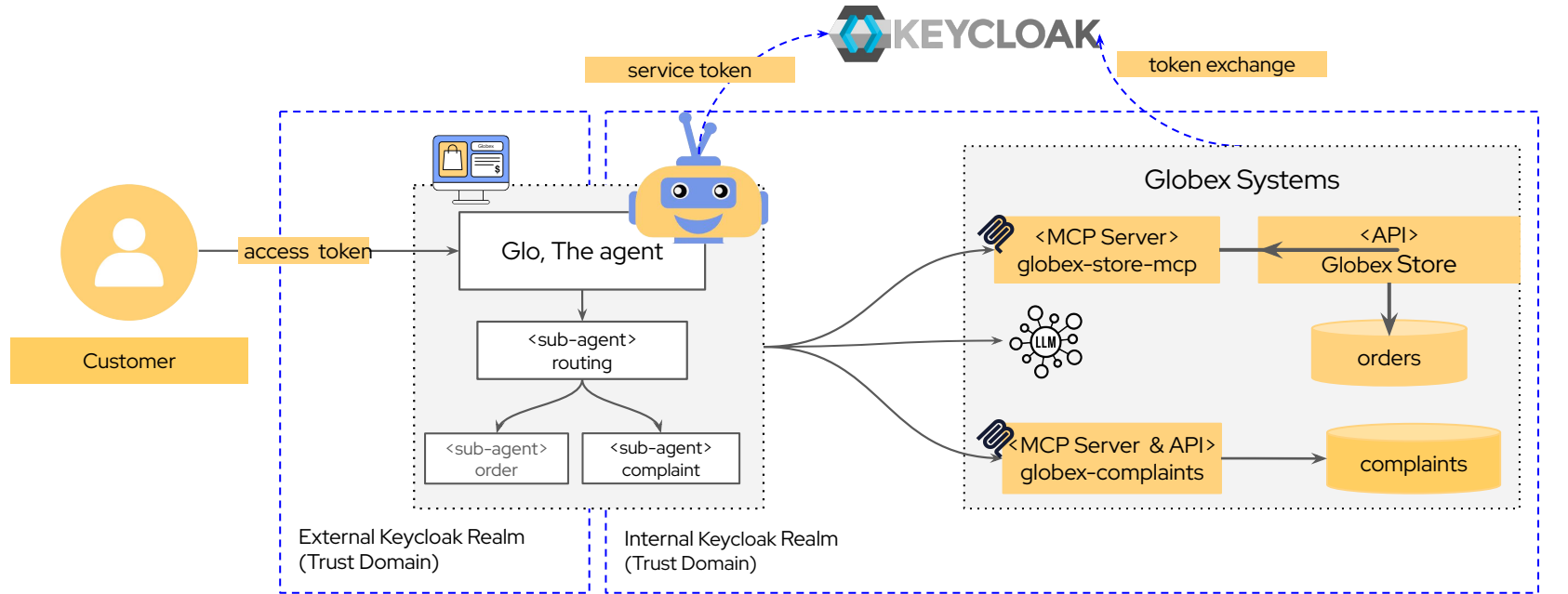
GET /orders/{id}
GET /orders/{id}/items
GET /orders/{id}/payment
POST /orders
PUT /orders/{id}

action!

[View API](#)



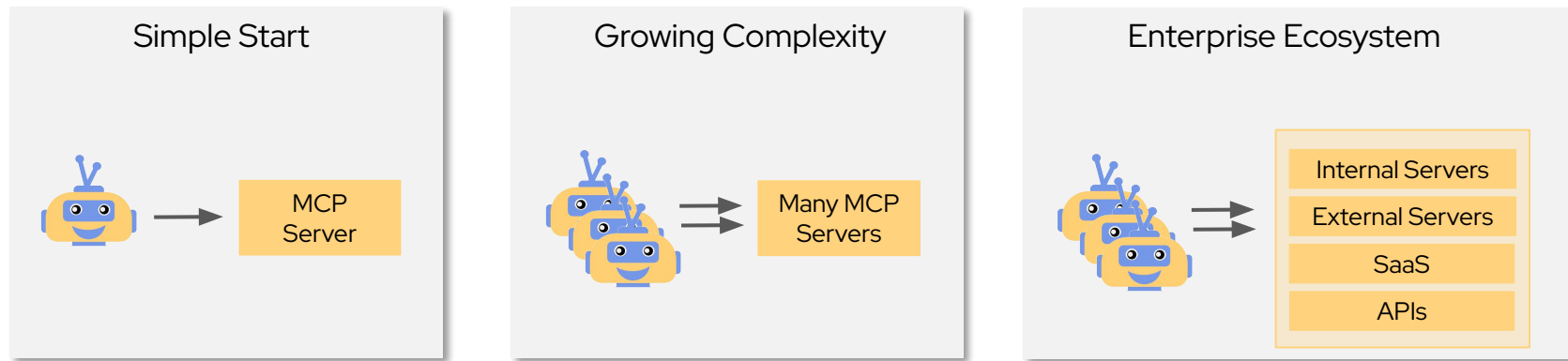
Securing the boundary



Raise Complaint >>



The MCP adoption curve



Single trust
boundary

Increasing Operational and
Governance Complexity

Multiple trust boundaries
Auth fragmentation
Tool sprawl
Observability gaps



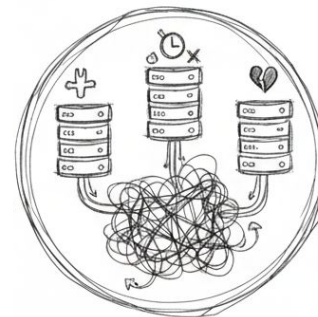
MCP has gained rapid adoption, but...



Uneven
evolution



Fragmented
protocol patterns

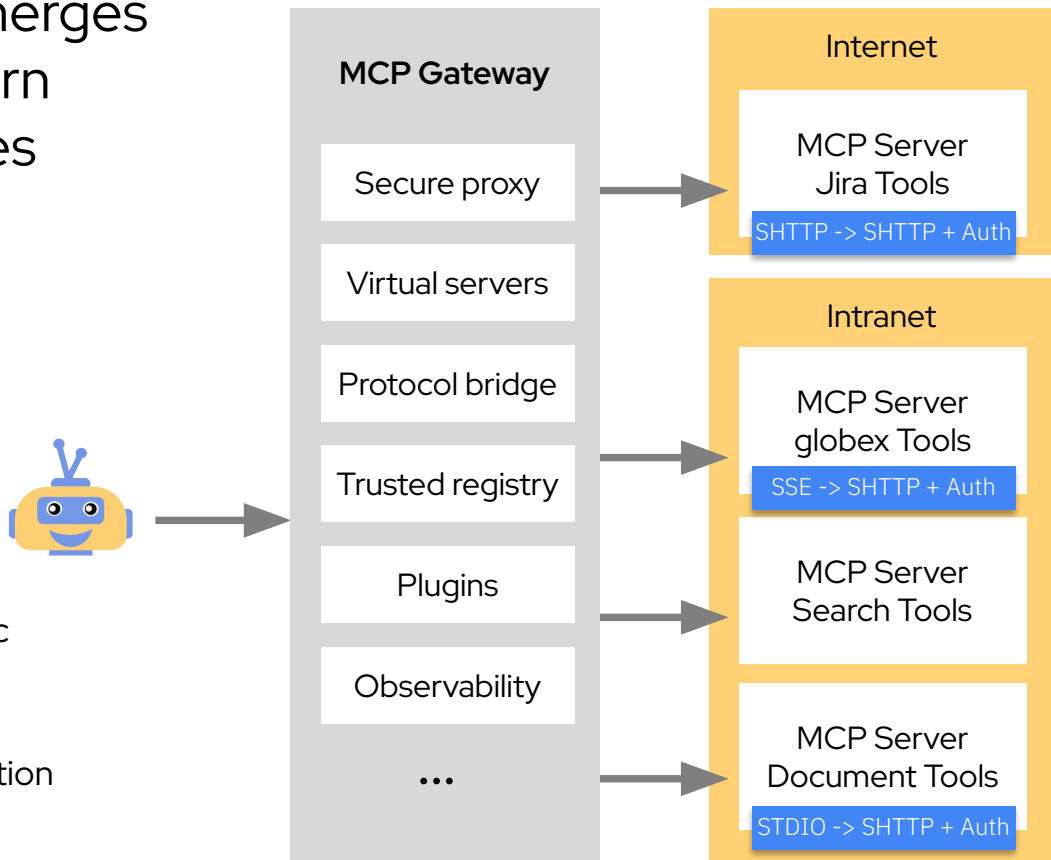


Integration
complexity



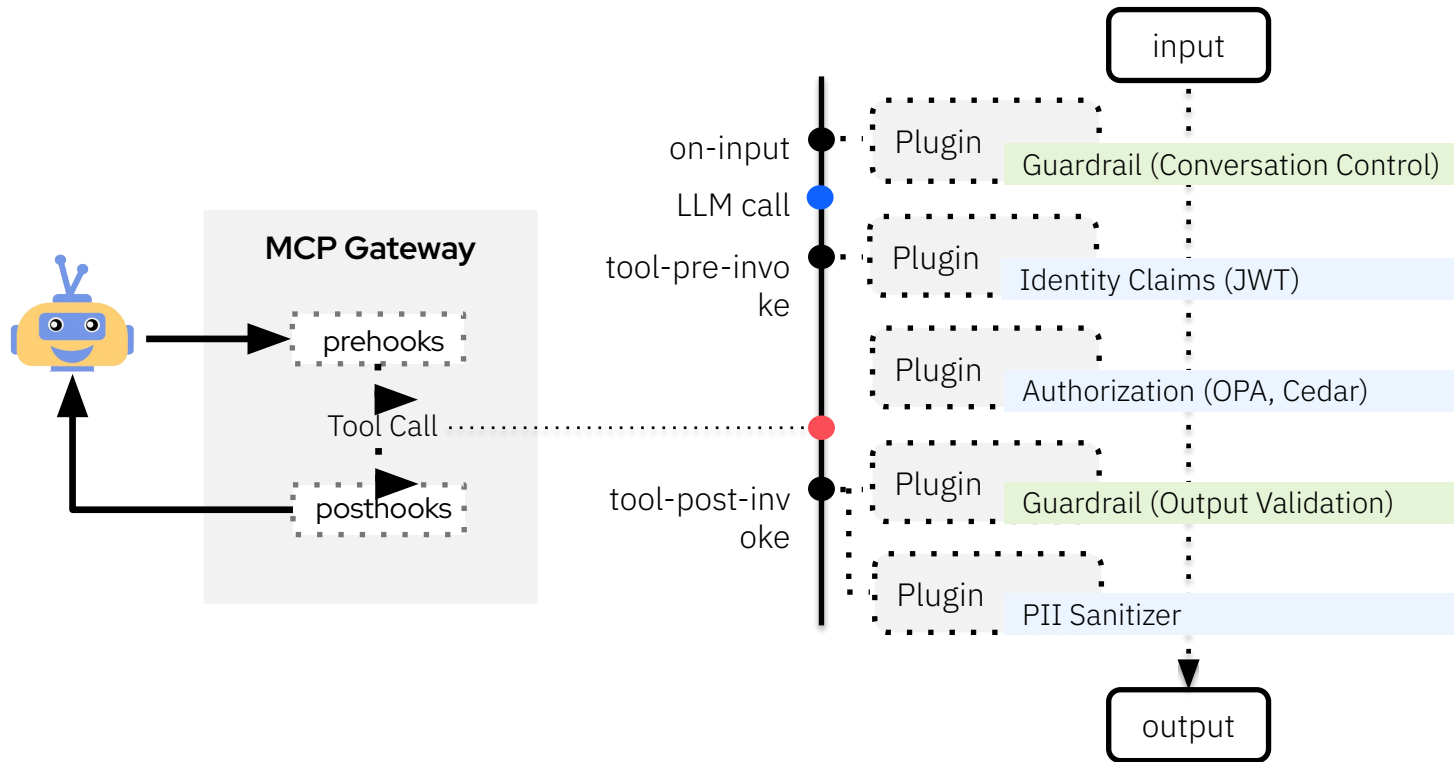
The MCP gateway emerges as an enterprise pattern to address these issues

- Unified entry point for MCP traffic
- Centralized policy enforcement
- Normalizes fragmented protocols
- Simplifies secure enterprise adoption
- Enables extensible controls



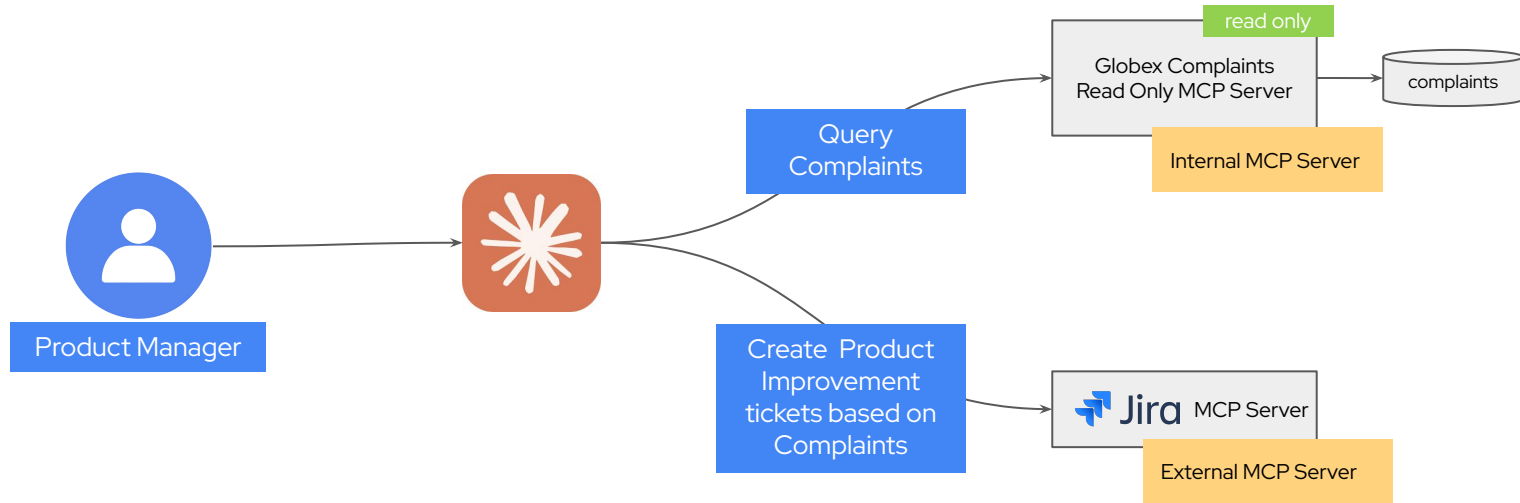
Extensible Security via Plugins

Composable governance for MCP ecosystems

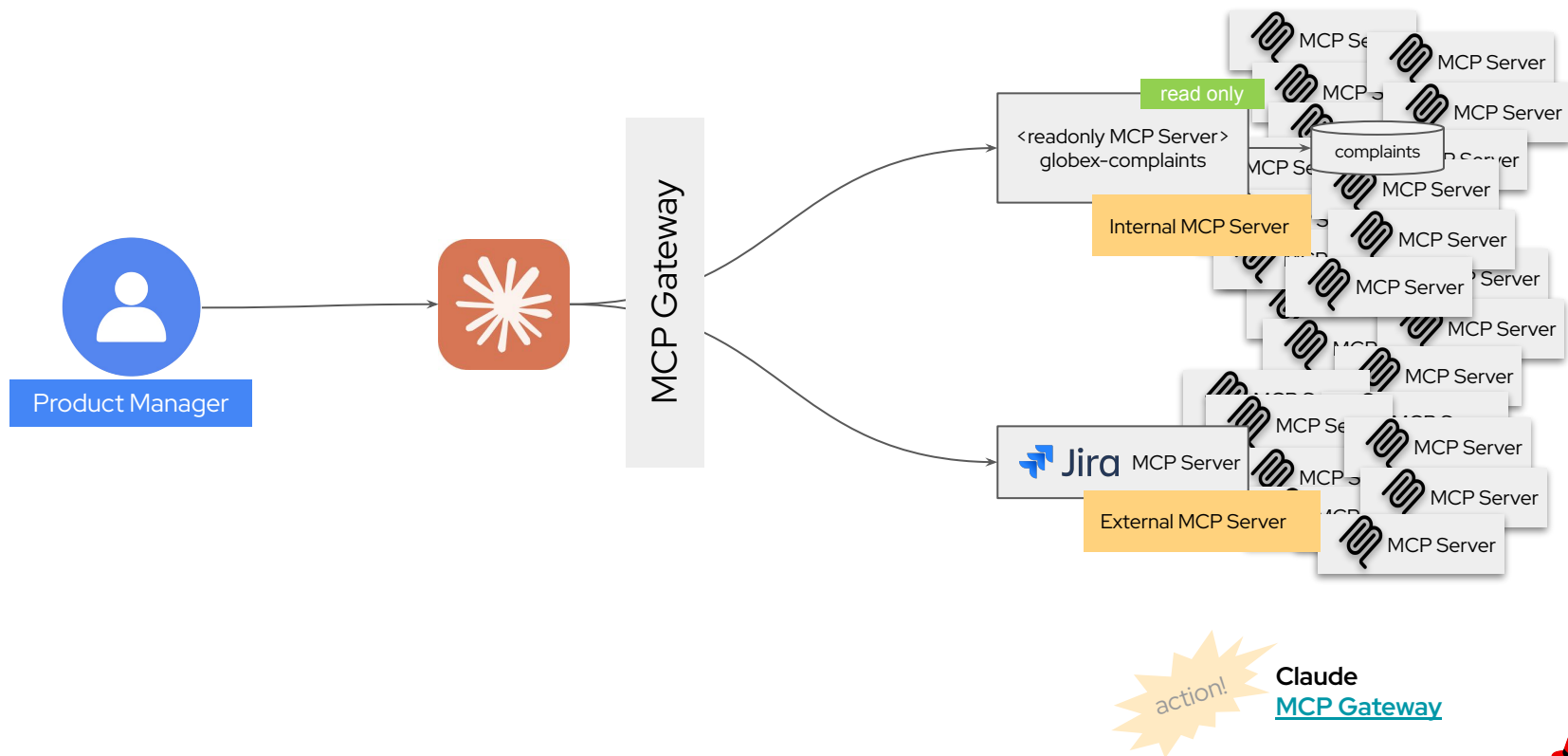


Business intelligence from Customer Complaints

Internal use case to derive Product Improvement ideas



MCP Gateway to govern and manage



Get the MCP Gateway to suit your environment

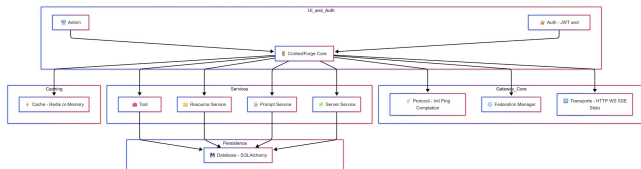
Open Source



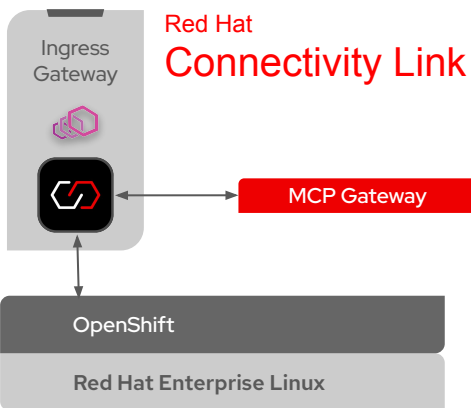
Red Hat



ContextForge AI Gateway



<https://github.com/ibm/mcp-context-forge>



Feb 2026 Developer preview
Apr 2026 Technical preview
Q4 2026 General availability



IBM DataPower Interact Gateway

Coming June 2026 as a standalone product, delivering dedicated, production-grade AI interaction governance.



What's next

- Demo source [red.ht/quarkus-mcp-demo](https://github.com/red-hat-quarkus/quarkus-mcp-demo)
- Context Forge:
<https://github.com/ibm/mcp-context-forge>
- Kuadrant MCP Gateway:
<https://github.com/Kuadrant/mcp-gateway>



red.ht/quarkus-mcp-demo





Thank you



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